

WEB TECHNOLOGIES

CSE-4004

LECTURE 1: HTML5- LISTS, FORM, FRAMESET

By: Dr.MD.Sirajuddin

LIST

- **Used to organize and display contents in a structured way.**
- **3-Types of Lists**
 - 1. Ordered List:** Used when sequence of items is important.
 - 2. Unordered List:** Used when order of items is not important.
 - 3. Description List:** Used to present items and their description (like FAQs, Glossary etc).

ORDERED LIST

```
<ol Type= " " >  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ol>
```

Example:

```
<ol type="1">  
  <li>Faculty</li>  
  <li>Student</li>  
  <li>Parents</li>  
</ol>
```

1. Faculty
2. Student
3. Parents

UNORDERED LIST

```
<ul>  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ul>
```

Example:

```
<ul>  
  <li>Faculty</li>  
  <li>Student</li>  
  <li>Parents</li>  
</ul>
```

- Faculty
- Student
- Parents

Default: disc

```
<ul type=  
"square">
```

DESCRIPTION LIST

```
<dl>  
  <dt>Term 1</dt>  
  <dd>Description</dd>  
  <dt>Term 2</dt>  
  <dd>Description</dd>  
</dl>
```

Example:

```
<dl>  
  <dt>WT</dt>  
  <dd>Web Technologies</dd>  
  <dt>HTML</dt>  
  <dd>Hyper Text Markup Language  
  </dd>  
</dl>
```

WT

Web Technologies

HTML

Hyper Text Markup Language

Table Tag

- It is used to organize data in form of table.

<table>: Defines table

<tr>: Table Row

<th>: Table Heading – Used for Column Headings

<td>: Table Data

<caption>: Table Title

Syntax

```
<table>
```

```
<tr>
```

```
<th>Column 1</th>
```

```
<th>Column 2</th>
```

```
<th>Column 3</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Data 1</td>
```

```
<td>Data 2</td>
```

```
<td>Data 3</td>
```

```
</tr>
```

```
</table>
```

```
<table border="1">
```

```
<tr>
```

```
<th rowspan="2">Name</th>
```

```
<th colspan="2">Marks</th>
```

```
</tr>
```

```
<tr>
```

```
<th>Maths</th>
```

```
<th>Science</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Ravi</td>
```

```
<td>85</td>
```

```
<td>90</td>
```

```
</tr>
```

```
</table>
```

Name	Marks	
	Maths	Science
Ravi	85	90

Form Tag

- Used to collect information from the user.
- A form sends the collected data to a server for processing.

```
<form action="submit.php" method="post">
```

```
<!-- form elements here -->
```

```
</form>
```

- **Form elements:** textbox, textarea, checkbox, button, drop-down list etc.

Registration Form

Name:

Message:

Country:

Languages Known: ☐ English ☐ Hindi

Male ☒ Female ☐

<h2>Registration Form</h2>

<form>

<label for="name">Name:</label>

<input type="text" id="name" name="username">

<label for="msg">Message:</label>

<textarea id="msg" name="msg"></textarea>

<label for="country">Country:</label>

<select id="country" name="country">

<option value="india">India</option>

<option value="usa">USA</option>

</select>

Registration Form

Name:

Message:

Country:

Languages Known: ☐ English ☐ Hindi

Male ☒ Female ☐

Note: clicking on the text Name: will automatically focus the textbox.


```

<label for="male">Male</label>
<input type="radio" id="male" name="gender"
value="male">
<label for="female">Female</label>
<input type="radio" id="female" name="gender"
value="female">
<label >Languages Known:</label>
<input type="checkbox" id="eng" name="lang"
value="English">
<label for="eng">English</label>
<input type="checkbox" id="hin" name="lang"
value="Hindi">
<label for="eng">Hindi</label>
<button type="submit">Submit</button> </form>

```

Registration Form

Name:

Message:

Country:

Languages Known: ☐ English ☐ Hindi

Male ☒ Female ☐

Note: clicking on the text Name: will automatically focus the textbox.

GET

- Sends form data in URL (after ?)
- Data is visible in address bar
- Not Secure
- Sends limited data
- *<https://example.com/search?name=John&passwd=25>*

POST

- Sends data inside the HTTP body, not in URL
- Data is not visible in address bar
- More Secure than GET
- Allows large amount of data.

EVENT

- It is an action that occurs in the browser.
- Triggered by user interaction or by the browser itself.
- Allows webpage to respond to the user and browser actions.

User Interaction Events

- Mouse events, Keyboard events, Form events.

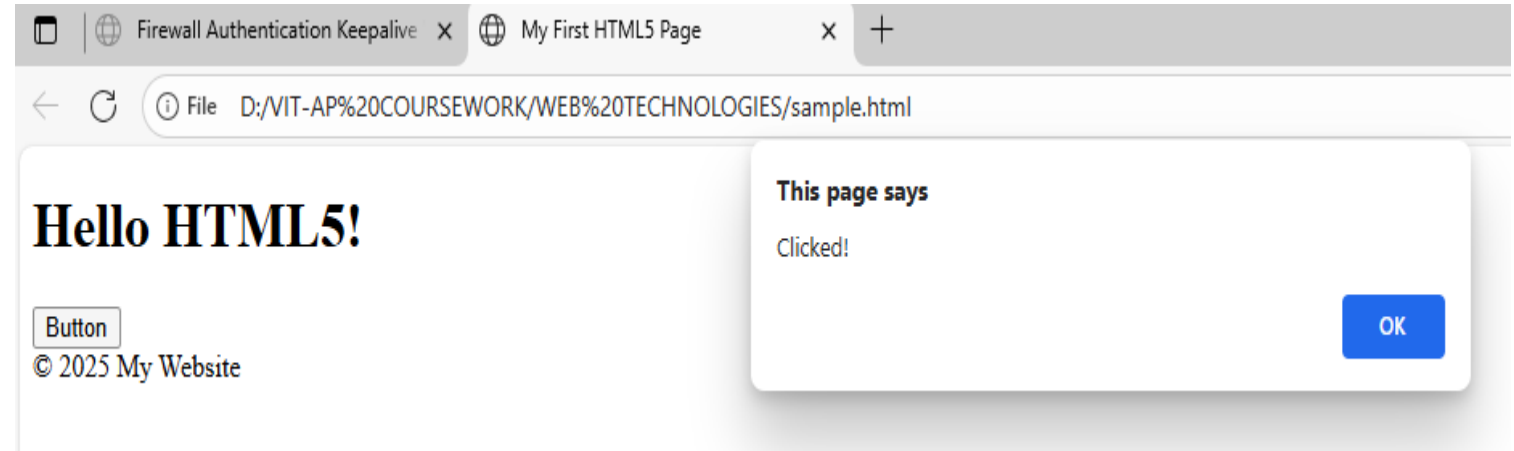
Browser Events

- onload, onresize, onunload etc.
- `<body onload="alert('page loaded')">`

MOUSE EVENTS

- **onclick**
- **ondblclick**
- **onmousedown**
- **onmouseup**
- **onmousemove**
- **onmouseover**
- **onmouseout**
- **onwheel**

**<button
onclick="alert('Clicked!')">Button</button>**



MOUSE EVENTS

- onkeyup
- onkeydown: fires for all keys
- Onkeypress: fires only for printable characters

FORM EVENTS

- onfocus
- onblur
- onchange
- onsubmit
- onreset

WINDOW EVENTS

- onload
- onunload
- onresize
- onscroll

SCALABLE VECTOR GRAPHICS (SVG)

- Used to draw vector graphics – shapes, charts, animations and icons.
- SVG graphics do not pixelate when zoomed

Example

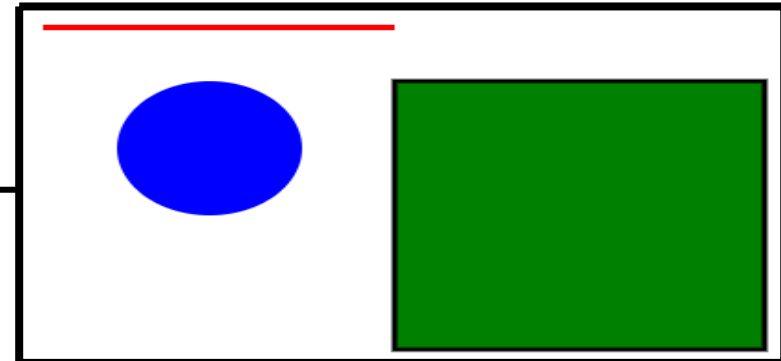
`<svg width="500" height="500">` *Creates canvas in the browser*

`<circle cx="100" cy="100" r="50" fill="blue" />`

`<rect x=200 y=50 width="200" height="200" fill="green" stroke="black" stroke-width="3"/>`

`<line x1="10" y1="10" x2="200" y2="10" stroke="red" stroke-width="4" />`

`</svg>`



Example: Ellipse

<ellipse cx="150" cy="100" rx="80" ry="40"/>

cx → x coordinate of the center

cy → y coordinate of the center

rx → Radius on the X-axis

ry → Radius on the Y-axis

SVG:ANIMATION

```
<svg width="500" height="500">
```

```
<circle cx="50" cy="100" r="30" fill="blue">
```

```
<animate
```

```
  attributeName="cx"
```

```
  from="50"
```

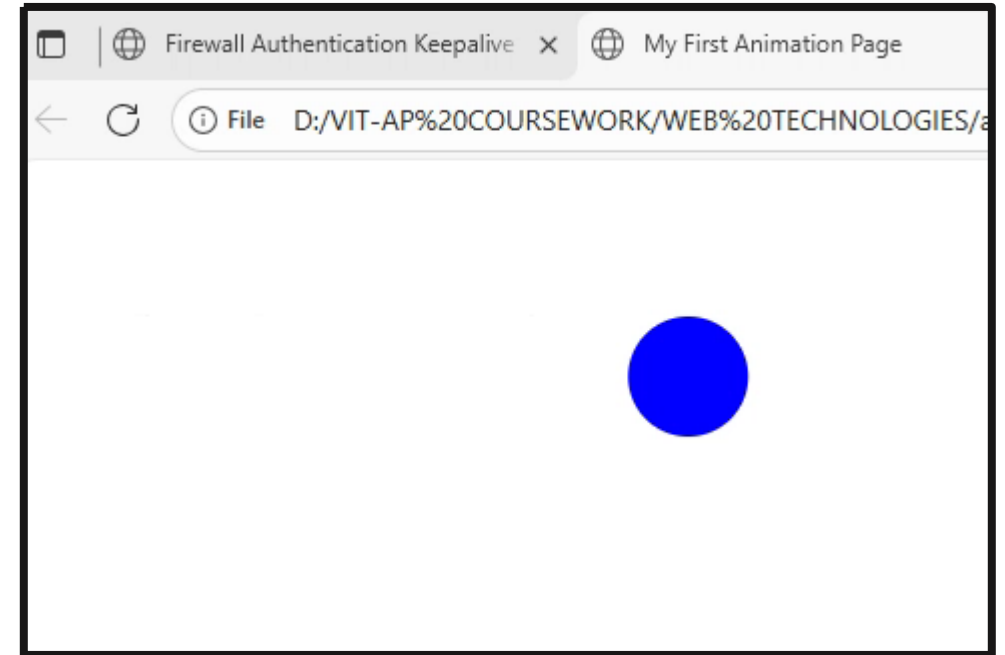
```
  to="350"
```

```
  dur="2s"
```

```
  repeatCount="indefinite" />
```

```
</circle>
```

```
</svg>
```



SVG:ANIMATION

```
<svg width="500" height="500">
```

```
<circle cx="200" cy="100" r="10" fill="red">
```

```
<animateTransform
```

```
  attributeName="transform"
```

```
  type="rotate"
```

```
  from="0 200 200"
```

```
  to="360 200 200"
```

```
  dur="4s"
```

```
  repeatCount="indefinite" />
```

```
</circle>
```

```
</svg>
```

SVG:ANIMATION

Transformation	Description
Rotate	Rotates an object about a fixed point transform="rotate(45 40 40)"
Scale	Increases or decreases object size transform="scale(1.5)" transform="scale(2,1)" Scale>1:Increase, Scale=1:Original, scale<1:decrease
Translate	Moves an object along X and Y axes transform="translate(50 10)"
Skew	Slants or Tilts an object along X or Y axis Eg: Rectangle → Parellelogram transform="skewX(30)"